

Last name _____

First name _____

LARSON—MATH 310—CLASSROOM WORKSHEET 07
Distance, Triangle Inequality, Standard Deviation.

Review

1. The **norm** of a vector $\vec{x} = (x_1, x_2, \dots, x_n)$ is $\|\vec{x}\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$.
2. The **root-mean square** of a n -vector $\vec{v}$ is $rms(\vec{v}) = \frac{\|\vec{v}\|}{\sqrt{n}}$.
3. Find $\vec{1}^T \vec{v}$ for $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$.
4. Find $\vec{v}^T \vec{1}$ for $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$.
5. Find $avg(\vec{v})$ for $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$.
6. Find the *de-meanned* vector $\tilde{v}$ for $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$.
7. The *standard deviation* of an n -vector $\vec{v} = rms(\tilde{v}) = \frac{\|\tilde{v}\|}{\sqrt{n}}$.

Norm and Distance (Chp. 3)

1. Let $\vec{a} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$. Check that $|\vec{a}^T \vec{b}| \leq \|\vec{a}\| \cdot \|\vec{b}\|$.
2. Find the angle $\angle(\vec{a}, \vec{b})$ between vectors $\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

3. Find the correlation coefficient ρ for vectors $\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

Linear Independence (Chp. 5)

4. What is the definition of the *linear dependence* of vectors $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_k$?
5. Check that vectors $\vec{a}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\vec{a}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\vec{a}_3 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are linearly dependent.
6. Check that one of $\vec{a}_1, \vec{a}_2, \vec{a}_3$ can be written in terms of the others.